PBL PROJECT PRESENTATION • 2026–27

FleetX

Intelligent Fleet & Logistics Management System

A unified platform for fleet operations, trips, shipments and tracking

PRESENTED BY

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Project Overview

FleetX brings key fleet and logistics workflows into one web application.

Fleet Operations

Manage vehicles, drivers, fuel records and maintenance.

Trip & Shipment Flow

Plan trips, assign resources and maintain shipment details.

Control Tower

Visualize vehicle movement, zones, alerts and operational status.

Project type: Java PBL Capstone • Team size: 5 • Academic year: 2026–27

Problem Statement

Fleet information can become fragmented across separate records and tools.

Coordinating vehicles, drivers, trips and shipments requires a shared operational view.

Manual status checks make it harder to spot delays, maintenance needs or route exceptions quickly.

A single system can improve visibility, consistency and traceability across the workflow.

Project Objectives

01 Centralize Data

Keep fleet, driver, trip and shipment records in one application.

02 Improve Visibility

Present status and route progress through maps and dashboards.

03 Support Decisions

Surface alerts, fuel efficiency and maintenance information.

04 Protect Access

Use session-based authentication, roles and CSRF protection.

Core Modules

Identity & Admin	Login, sessions, roles and user management
Fleet & Assets	Vehicle directory, fuel logs and maintenance
Drivers & Trips	Driver onboarding, dispatch and trip lifecycle
Shipments & Routes	Shipment manifests, waypoints and timelines
Tracking & Control Tower	GPS simulation, geofences, alerts and reports

User Roles & Access

Admin

Manage users and system-level access.

Fleet Manager

Oversee vehicles, assets and maintenance.

Dispatcher

Coordinate trips, drivers and shipment movement.

Driver

Access relevant trip and shipment information.

Viewer

View operational information with restricted permissions.

Access is enforced by the backend; UI visibility alone is not the security boundary.

System Architecture

FRONTEND • React 18 + TypeScript + Vite

Web UI | Leaflet maps | Three.js view | REST client | STOMP client

REST API ↓ / WebSocket events ↑

BACKEND • Java 21 + Spring Boot 3.5.9

Modular monolith: Auth | Fleet | Drivers & Trips | Shipments | Tracking

JPA / Hibernate ↓ • Flyway schema migrations

DATABASE • MySQL 8.4 LTS

Single deployable modular monolith — not a microservices system.

Technology Stack

Frontend

React 18 • TypeScript • Vite • Bootstrap 5.3

Maps & Visualization

Leaflet • OpenStreetMap tiles • Three.js

Backend

Java 21 • Spring Boot 3.5.9 • Spring Security

Persistence

MySQL 8.4 • Spring Data JPA • Hibernate • Flyway

Real-time & Build

WebSocket/STOMP • SockJS fallback • Maven • npm

Operational Workflow

1

Register

Maintain vehicle and driver records

→

2

Plan

Create trip and shipment details

→

3

Dispatch

Connect resources to planned operations

→

4

Track

Follow route waypoints and simulated GPS updates

→

5

Review

Check alerts, reports and operational history

One connected flow from resource setup to trip visibility and reporting.

Live Tracking & Control Tower

2D Map View

Leaflet map with OpenStreetMap tiles and route waypoints.

GPS Simulation

SimulatedGpsService interpolates coordinates along route waypoints.

Real-time Events

STOMP WebSocket events update tracking views and operational state.

Important disclosure: location telemetry is simulated and labeled “SIMULATED DATA”.

Route distance and ETA are estimates based on the Haversine straight-line calculation.

Fleet, Drivers & Maintenance

Vehicle directory stores identification, type, capacity and operational status.

Driver records support onboarding and relevant license details.

Trip lifecycle supports planned and in-progress operations.

Fuel logs calculate fuel-efficiency metrics such as kilometres per litre.

Maintenance records help teams review upcoming or preventive service needs.

Shipments & Route Planning

Shipment Manifest

Track shipment code, sender/receiver, weight, volume and cargo description.

Route Waypoints

Represent planned route points and display a shipment timeline.

Delivery Estimate

Show estimated straight-line distance and estimated arrival time.

The workflow connects shipment records to trips for a shared operational view.

Security & Data Management

Session-Based Authentication

Spring Security server-side sessions; role-based authorization.

CSRF Protection

State-changing requests require CSRF token forwarding.

Password Protection

BCrypt password hashing and secure session cookies.

Database Migrations

Flyway manages schema changes; Hibernate validates the schema.

MySQL uses a persistent Docker volume for local database storage.

Testing & Verification

Backend Tests

README describes 79 automated unit, module, migration and E2E tests.

Frontend Tests

Vitest + React Testing Library; README lists 7 suites and 34 tests.

Smoke Checks

Script checks health, SPA fallback, auth, CSRF, data access and logout.

Verification approach: validate each module, then test integrated workflows and security boundaries.

Run commands are documented in README.md: ./mvnw test, npm test -- --run, npm run build.

Benefits & Current Limitations

Benefits

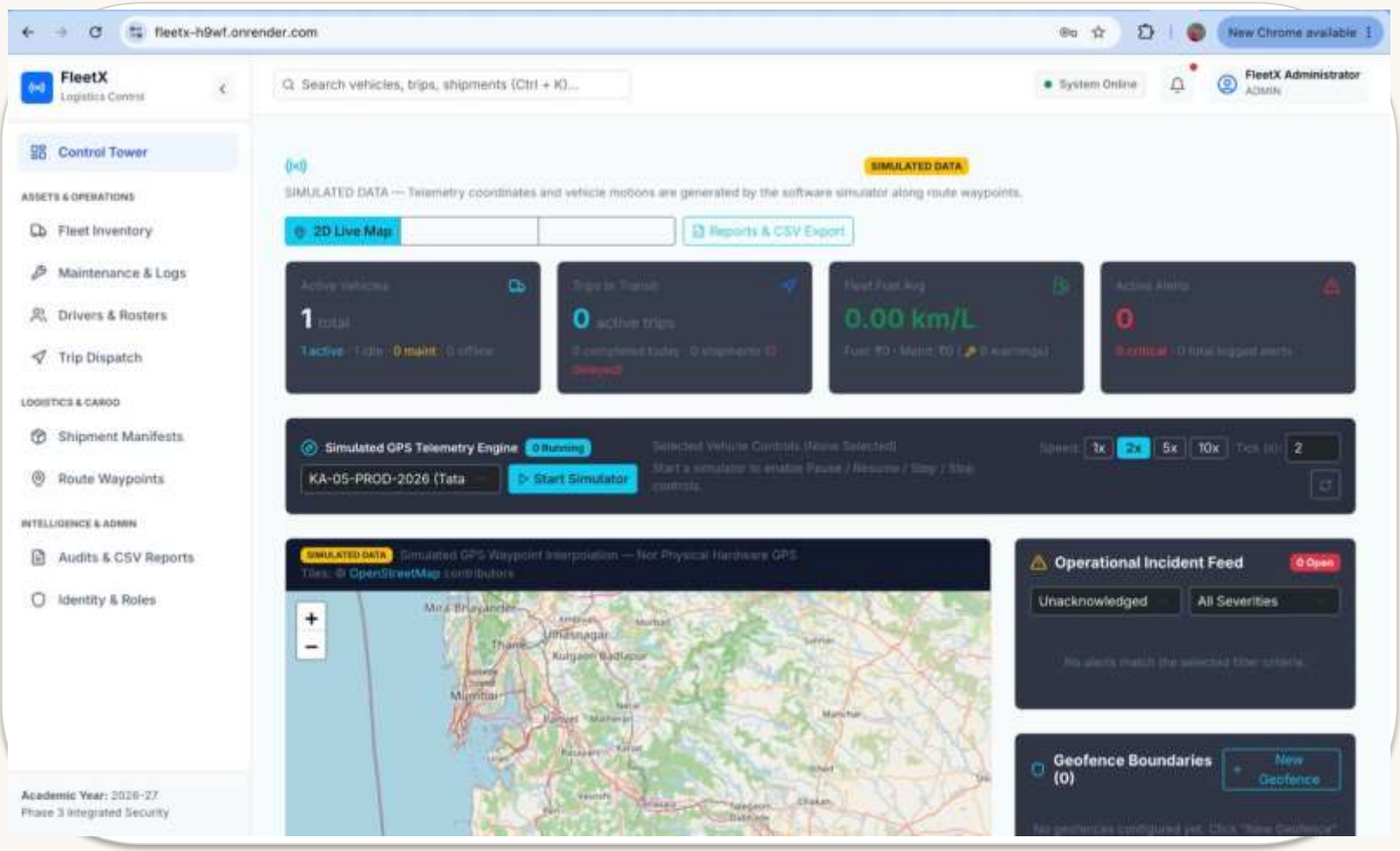
- Unified operational records
- Better trip and shipment visibility
- Role-aware access control
- Maintenance and fuel information
- Real-time-style tracking interface

Current Limitations

- GPS telemetry is simulated, not physical GPS hardware
- Haversine route length is straight-line, not road distance
- ETA is an estimate, not a traffic-aware prediction
- Production deployment requires environment-specific setup

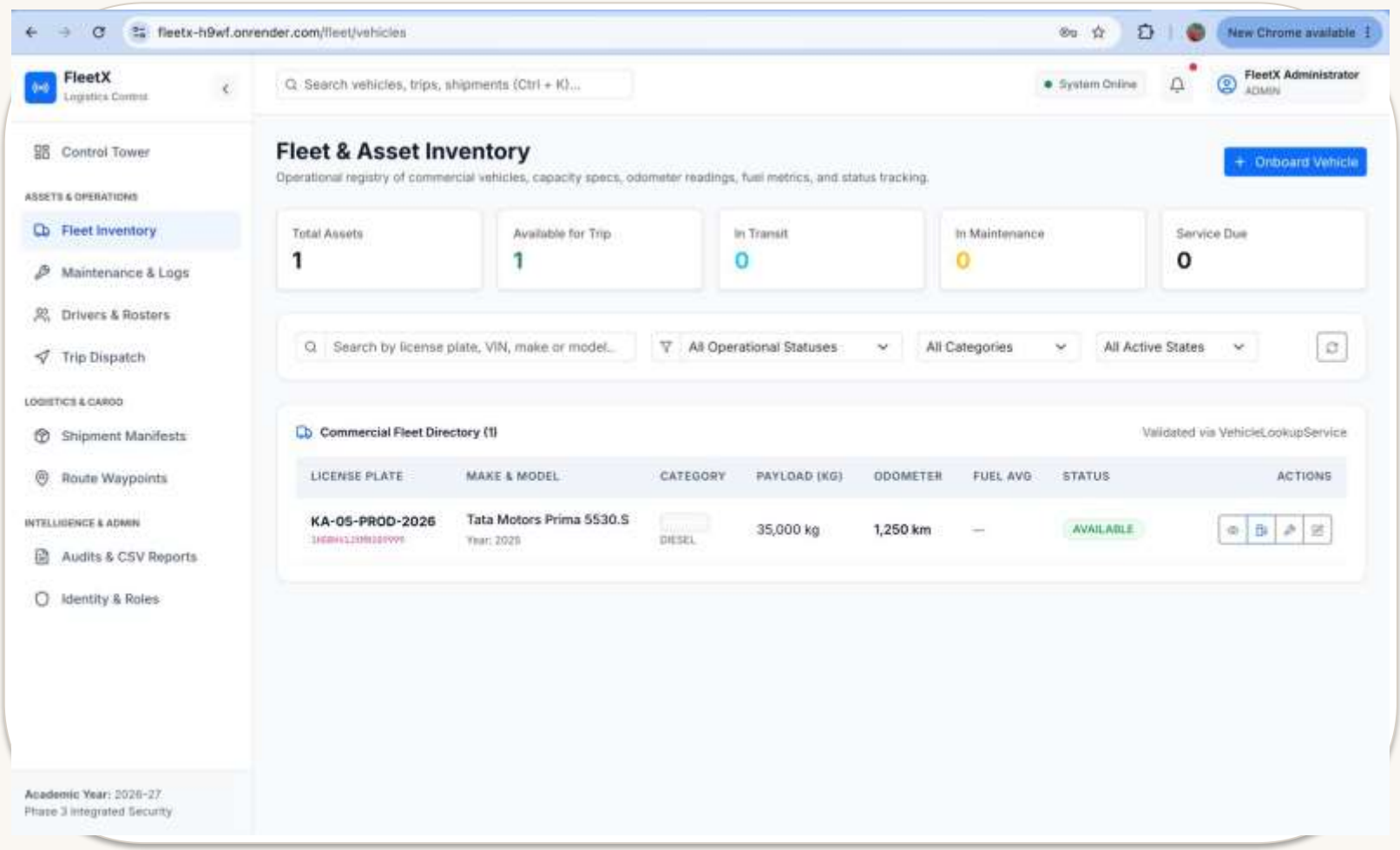
Clear disclosure of simulation and estimation keeps the project results academically transparent.

Control Tower & Simulated Live Tracking



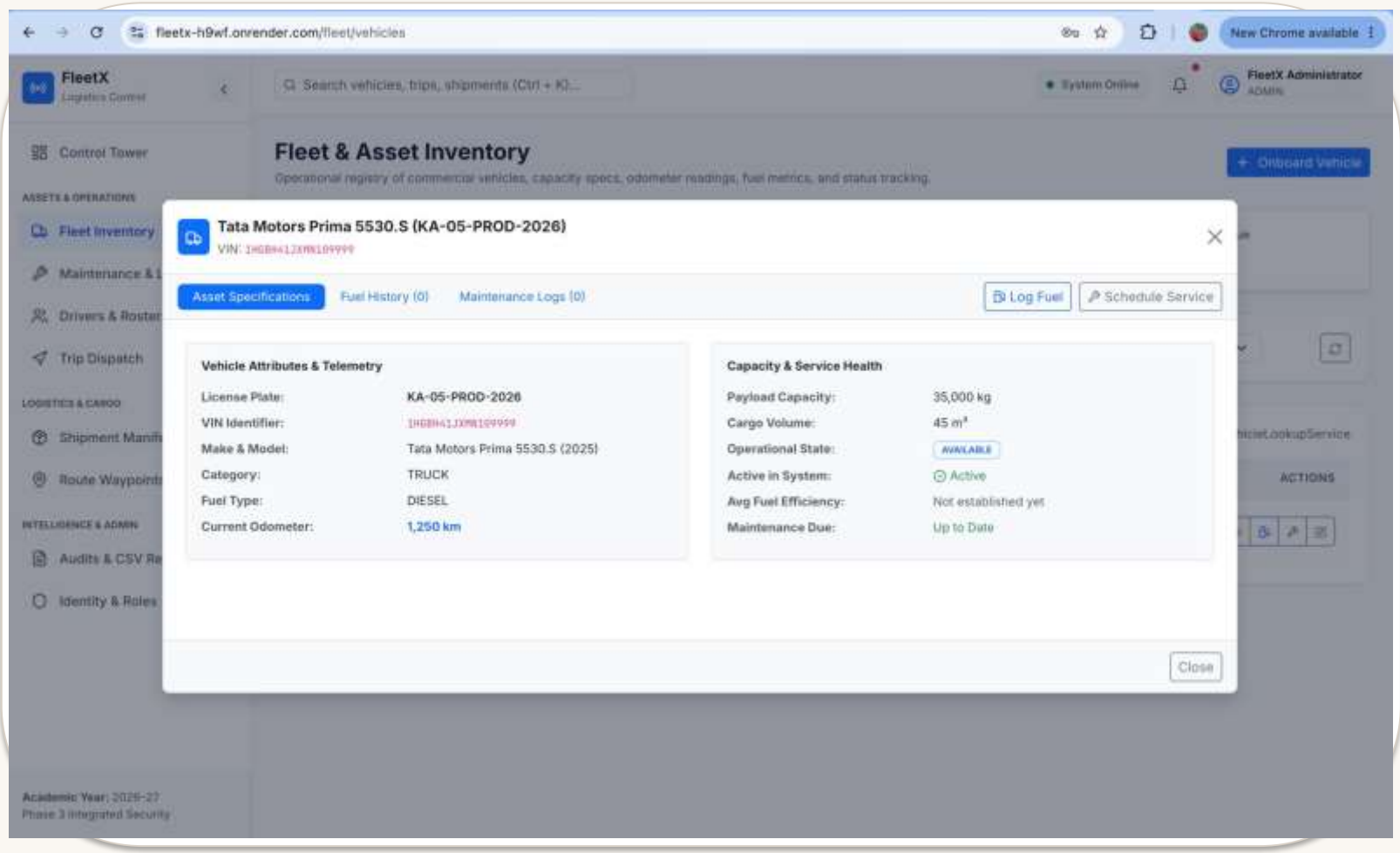
The Control Tower combines fleet KPIs, simulated GPS movement, map context, incident monitoring and geofence controls in one operational view.

Fleet & Asset Inventory



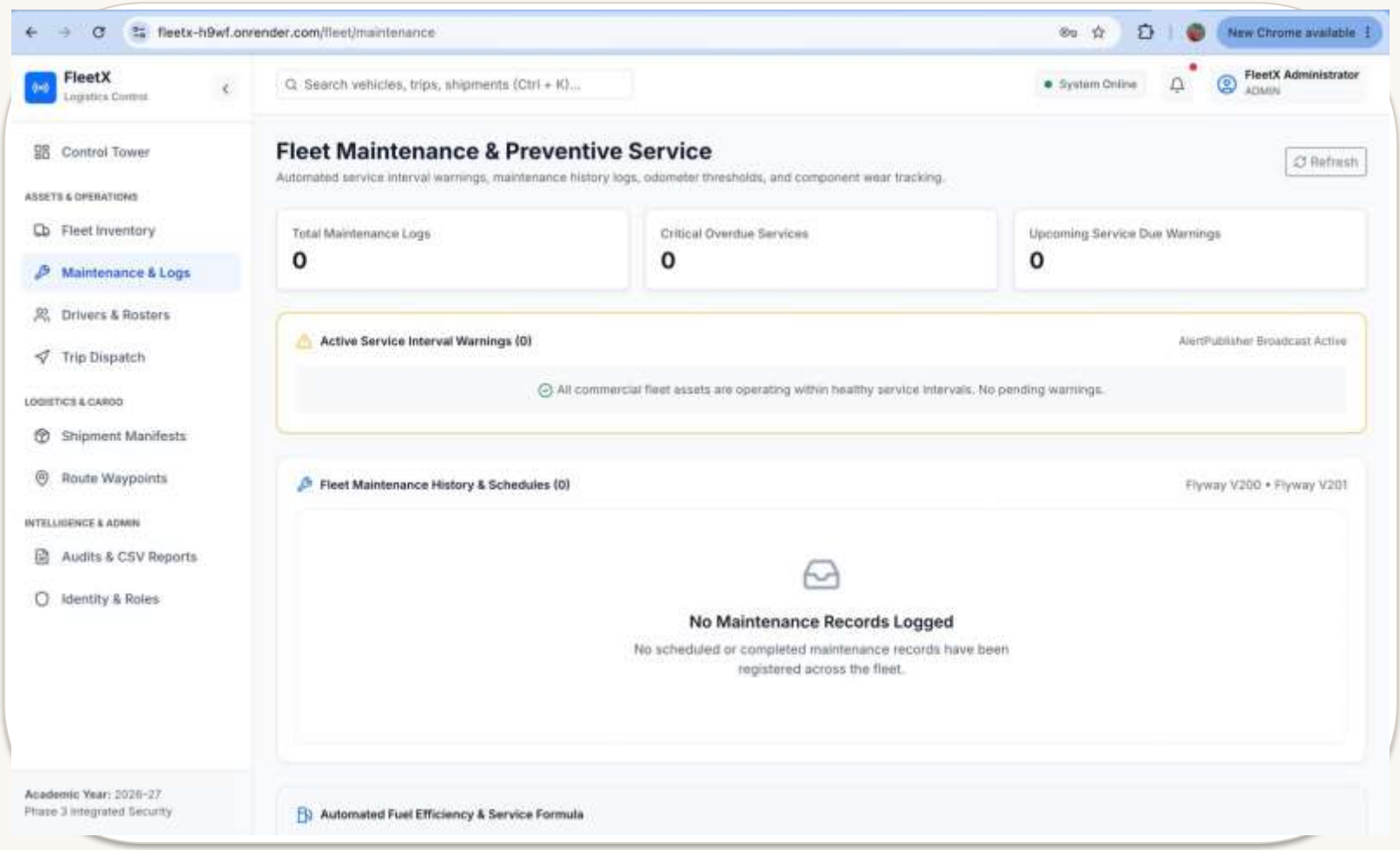
The fleet directory displays vehicle identity, category, payload capacity, odometer readings and current operational status.

Vehicle Specifications & Asset Actions



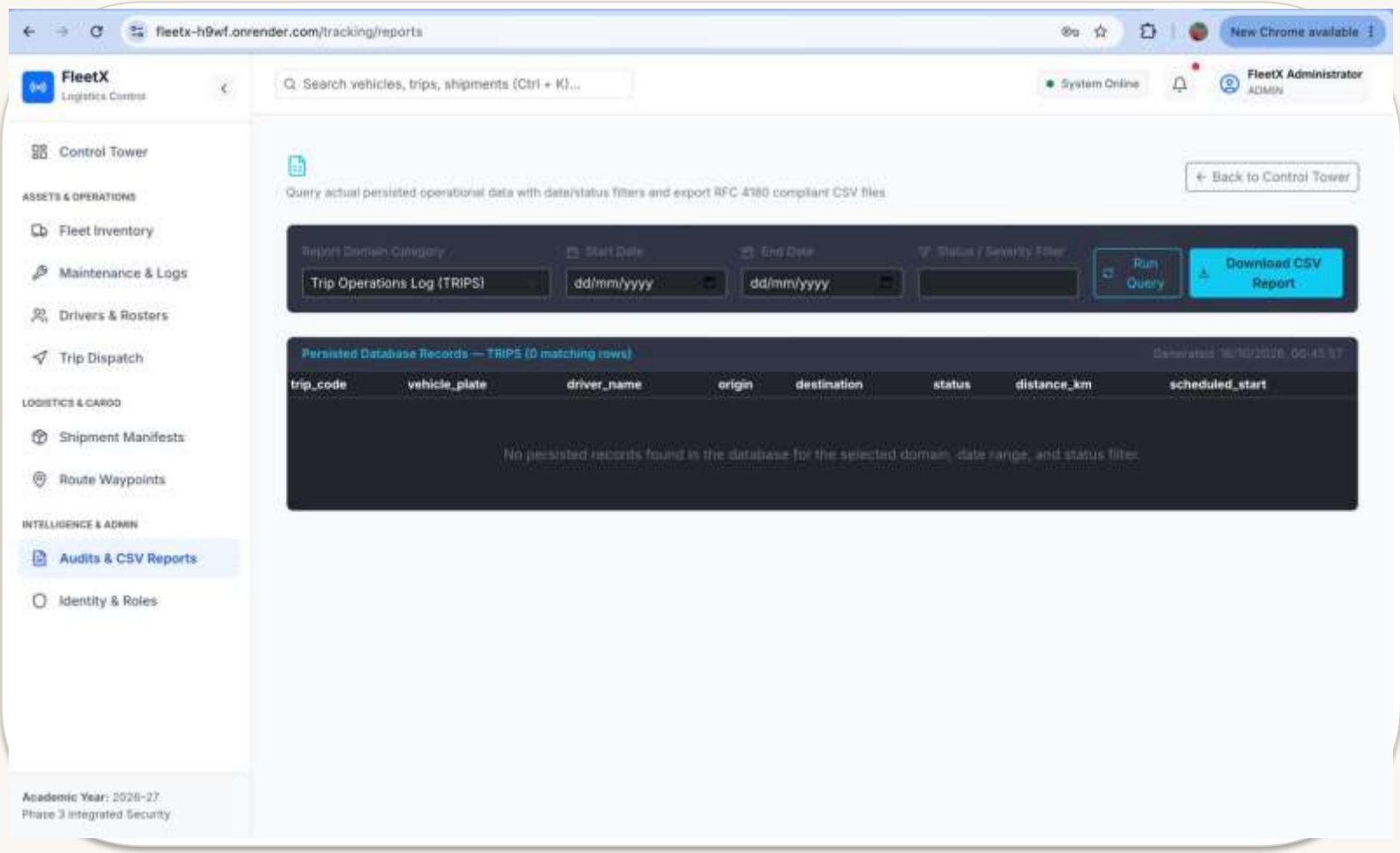
The vehicle details panel groups telemetry and capacity information with fuel history, maintenance logs, fuel entry and service scheduling actions.

Maintenance & Preventive Service



The maintenance page summarizes maintenance logs, overdue services, upcoming service warnings and maintenance history.

Audits, Operational Reports & CSV Export



The reporting page filters persisted trip records by domain, date range and status, with CSV export for further analysis.

One connected view of fleet operations.

FleetX combines fleet records, trip and shipment workflows, map-based tracking, alerts and reporting in a modular web application.

Thank You

Questions & Discussion